GG-B-476c
FEBRUARY 12, 1958
SUPERSEDING
Int. Fed. Spec. GG–B–00476b (GSA–FSS)
June 13, 1957, and
Fed. Spec. GG–B–476a
November 15, 1948

FEDERAL SPECIFICATION

BLOCK, SOLDERING, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers heat resistant blocks which serve as a support for soldering precious metals and cobalt-chromium alloys for dental appliances.

1.2 Classification.

1.2.1 *Types.*—Blocks shall be of the following types, as specified (see 6.2).

Type I.—Asbestos, soldering.

Type II.—Charcoal, soldering and melting.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–585—Boxes; Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington; D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

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FSC 6520

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MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Asbestos.* — Type I blocks shall be composed of asbestos fiber and suitable heat-resistant inorganic fillers, impregnated with an inorganic liquid binder, molded to shape in a cold plastic mass, and heat treated.

3.1.2 *Charcoal.* — Type II blocks shall be pure charcoal free from uncarbonized material and treated to retard combustion.

3.2 Dimensions, style, and design.

3.2.1 *Type I, asbestos, soldering.* — The block shall be round in shape with a flat bottom, 4½ inches in diameter and 1¾ inches in height, with a concavity in the center of block 3½ inches in diameter and 1 inch in depth. Depth of concavity shall have a tolerance of ± ⅛ inch. Tolerance for all other dimensions shall be ± ¼ inch.

3.2.2 *Type II, charcoal, soldering and melting.*—The block shall be rectangular in shape, 4¾ inches in length, 3 inches in width, and 1⅜ inches in height. Dimensional tolerance shall be ± ¼ inch.

3.3 **Heat resistance.** — Type II charcoal blocks shall withstand the heat-resistance test specified in 4.3.1 without any material damage, such as, cracking or disintegrating. The presence of a thin film of white ash on surface of block resulting from this test shall not be construed as a material defect.

3.4 **Workmanship.**—Workmanship shall be first class throughout. The blocks shall be smoothly finished and free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105. In addition, samples may be requested by the procuring activity for testing at Government laboratory.

4.2 **Inspection.** — Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery, as specified (see 6.2). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 Tests.

4.3.1 *Heat-resistance test.*—Charcoal block shall be placed in an air-circulating drying oven for 2 hours at 105° C. ± 2° C. before being subjected to the high heat of 1200° C. A muffle furnace shall be heated to 1200° C. ± 10° C. and when maintained at this temperature for 20 minutes, the dried block shall then be placed in the furnace for 10 minutes. At the end of the 10-minute test period, the block shall be removed and placed in a container with a loose cover and allowed to cool to 400° C. ± 10° C. The block is then removed and placed on a strip of asbestos paper in the open and allowed to cool to room temperature. The same test block shall be subjected to an additional 10-minute test at 1200° C. level followed by the same test procedure as for the first cycle. The block shall be examined for compliance with requirements of 3.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.*—One block, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue, indicated in table I, shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal Stock Number	Unit package	Intermediate package	Shipping container
6520–501–7850	1 unit	6 units	144 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be wrapped in kraft paper of suitable weight. Wrapping shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 *Level C.*—Unless otherwise specified, each block shall be packaged in accordance with accepted commercial practice.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Unit package.*—Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

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5.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2.1).

(c) Inspection required (see 4.2).

(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the title of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 This specification covers the following item appearing in the Armed Services Medical Stock List:

6520–501–7850—BLOCK, SOLDERING, DENTAL, Charcoal.

6.6 Transportation description.—Transportation description applicable to this item is:

Type I:

Asbestos articles, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.
Motor volume minimum weight 30,000 pounds.

Type II:

Charcoal, wood, not otherwise indexed by name, not activated.
Carload minimum weight 26,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 26,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—M
Navy—Md.
Air Force.

Copies of this specification may be purchased for 5 cents each.

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